a de minimis system disruption) has occurred, a competing consolidator will be required to publicly disseminate information relating to the event; when known, promptly publicly disseminate additional information relating to the event; and until resolved, provide regular updates with respect to such information.[1587] Concurrent with public dissemination of information relating to a systems disruption, competing consolidators will also be required to provide the Commission notification of such event, including the information required to be publicly disseminated.[1588] In addition, competing consolidators will be required to notify the Commission upon responsible personnel having a reasonable basis to conclude that a systems intrusion (other than a de minimis system intrusion) has occurred. Notifications regarding systems disruptions and systems intrusions that competing consolidators must provide to the Commission under this provision include information relating to the event; when known, additional information relating to the event; and until resolved, regular updates with respect to such information. Rule 614(d)(9)(iv) will require competing consolidators to participate in the industry- or sector-wide coordinated testing of BC/DR plans required of SCI entities pursuant to paragraph (c) of Rule 1004 of Regulation SCI. The Commission believes this requirement will involve notifying market participants and scheduling the coordinated testing.

(b) Use of Information

Paragraph (d)(9)(ii) of Rule 614 should help to advance the goal of promoting Commission review and oversight of market data infrastructure by requiring a competing consolidator to have policies and procedures that are reasonably designed to ensure its operational capability, including the ability to maintain effective operations; minimize or

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1587 See Rule 614(d)(9)(iii)(B) of Regulation NMS.

1588 See Rule 614(d)(9)(iii)(C) of Regulation NMS.

eliminate the effect of performance degradations; and help ensure the prompt, accurate, and reliable dissemination of consolidated market data. Because a competing consolidator's operational capability can have the potential to impact market participants who rely on such competing consolidators for market data, the Commission believes that these policies and procedures will help promote the maintenance of fair and orderly markets.

The requirement in paragraph (ii)(C) of Rule 614(d)(9) to establish policies and procedures that include the designation and documentation of responsible personnel should help make it clear to all employees of the competing consolidator who the designated responsible personnel are for purposes of the escalation procedures and so that Commission staff can easily identify such responsible personnel in the course of its inspections and examinations and other interactions with competing consolidators. The Commission also believes that escalation procedures to quickly inform responsible personnel of potential systems disruptions and systems intrusions helps ensure that the appropriate person(s) are provided notice of potential systems issues so that any appropriate actions can be taken in accordance with the requirements of Rule 614(d)(9) without unnecessary delay.

Rule 614(d)(9)(iii)(A) should help facilitate competing consolidators' responses to systems disruptions and systems intrusions, including taking appropriate steps necessary to remedy the problem or problems causing such event and mitigate the negative effects of the event, if any, on market participants and the securities markets more broadly.

Rule 614(d)(9)(iii)(B) should help to advance the Commission's goal of promoting fair and orderly markets by publicly disseminating information about systems disruptions, allowing market participants to use such information to evaluate the event's impact on their trading and

other activities and develop an appropriate response, as well as to evaluate the performance of various competing consolidators.

Rule 614(d)(9)(iii)(C) provides for a framework for reporting of systems disruptions and systems intrusions, which ensures the Commission's review and oversight of market data infrastructure and fosters cooperation between the Commission and competing consolidators in responding to such events. The Commission also believes that the aggregated data from the reporting of systems disruptions and systems intrusions, in combination with filings from SCI competing consolidators under Regulation SCI, enhances its ability to comprehensively analyze the nature and types of various systems issues and identify more effectively areas of persistent or recurring problems across the systems of all competing consolidators.

Rule 614(d)(9)(iv) should assist the Commission in maintaining fair and orderly markets in a BC/DR scenario following a wide-scale disruption.

(c) Collection of Information is Mandatory

The collection of information discussed above is a mandatory collection of information.

(d) Confidentiality

The Commission expects that the written policies and procedures, processes, criteria, standards, or other written documents developed or revised by competing consolidators pursuant to Rule 614(d)(9) will be retained by competing consolidators in accordance with, and for the periods specified in, applicable recordkeeping requirements. Should such documents be made available for examination or inspection by the Commission and its representatives, they would be kept confidential subject to the provisions of applicable law. In addition, the information submitted to the Commission that is filed on Form CC is public, as discussed in detail above.

The information publicly disseminated by competing consolidators pursuant to Rule 614(d)(9)(iii)(B) is not confidential.

(e) Respondents

As described above, the Commission estimates that, following the SCI CC Phase-In Period, one of the eight competing consolidators will not meet the definition of "SCI competing consolidator" and will be subject to the requirements of subparagraph (d)(9) of Rule 619.

(f) Total Initial and Annual Reporting and Recordkeeping Burden

As described in detail above, the requirements under Rule 614(d)(9) are substantially similar to a subset of the requirements of Regulation SCI. In particular, these provisions largely mirror the requirements of Regulation SCI Rules 1001(a)(1), 1001(a)(2)(vi), 1001(a)(4), 1001(a)(3), 1001(c), 1002(a), 1002(c), and 1004(c). Accordingly, the Commission believes that its 2018 burden estimates for these rules would be applicable to the corresponding requirements under Rule 619(d)(9). With regard to the Commission notification provision in subparagraph (d)(9)(ii)(C), as described above, the Commission believes that this provision is significantly more streamlined than the requirements under Rule 1002(b), and therefore competing consolidators would incur only a small portion of the estimated burdens for Rule 1002(b). Considering its prior burden estimates for the Regulation SCI rules, the Commission estimates that the one competing consolidator subject to the requirements of Rule 619(d)(9) following the SCI CC Phase-In Period will have initial and ongoing burdens that are approximately 33% of the burdens estimated for compliance with all of the provisions of Regulation SCI.[1589] This estimate of 33% includes the paperwork burdens estimated for Rules 1001(a)(1), 1001(a)(2)(vi), 1001(a)(4), 1001(a)(3), 1001(c), 1002(a), 1002(c), and 1004(c) of Regulation SCI, with the

[1589] See 2018 PRA Extension, supra note 1567.

addition of an incremental burden associated with notifying the Commission of systems disruptions and systems intrusions on Form CC, as compared to the burden estimates for all of the requirements of Regulation SCI that will be applicable to SCI competing consolidators.

V. Economic Analysis

A. Introduction and Market Failures

1. Introduction

Section 3(f) of the Exchange Act requires the Commission, whenever it engages in rulemaking and is required to consider or determine whether an action is necessary or appropriate in the public interest, to consider, in addition to the protection of investors, whether the action would promote efficiency, competition, and capital formation.[1590] In addition, Section 23(a)(2) of the Exchange Act requires the Commission, when making rules under the Exchange Act, to consider the impact such rules would have on competition.[1591] Exchange Act Section 23(a)(2) prohibits the Commission from adopting any rule that would impose a burden on competition not necessary or appropriate in furtherance of the purposes of the Exchange Act.

The Commission believes that the economic benefits of the amendments justify the costs. The amendments will generally enhance the consolidated market data content, reduce the latency of consolidated market data, and improve the dissemination of consolidated market data. This will reduce information asymmetries that exist between market participants who subscribe to proprietary DOB and other proprietary products and market participants who only subscribe to SIP data, and may allow some market participants who subscribe to proprietary DOB products to replace them with potentially cheaper consolidated market data feeds. Improvements to the

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[1590] 15 U.S.C. 78c(f).

[1591] 15 U.S.C. 78w(a)(2).

content and latency of consolidated market data from the amendments may also help market participants that currently rely on SIP data to make more informed trading decisions, which will facilitate their ability to trade competitively and improve their execution quality, and will facilitate best execution.

The Commission perceives three main benefits from the new round lot definition and the expanded content of consolidated market data, which as noted above includes "core data." First, the expanded content of consolidated market data will enable market participants[1592] that currently only subscribe to SIP data to get additional content from expanded consolidated market data and to experience increased gains from trade by allowing them to take advantage of trading opportunities they may not have been aware of due to the lack of information in existing SIP data.[1593] Second, the expanded content of consolidated market data may also allow these market participants to make more informed trading decisions and improve their order routing and order execution capabilities, potentially lowering investor transaction costs. Finally, the changes in the definition of the round lot will result in a narrower NBBO in some higher priced stocks, which may improve execution quality. A narrower NBBO could also affect the amount of price improvement that trading venues, including ATSs, exchanges, and internalizers could offer.

[1592] Here, market participants may include investors, including retail investors. Market participants that do not receive the additional content from expanded consolidated market data may benefit indirectly if the broker-dealers that handle their orders subscribe to the expanded content. The extent to which particular kinds of market participants will incur benefits or costs from these final rules is discussed more fully in the relevant parts of Section V.C.

[1593] Here and throughout, the phrase "gains from trade" refers to a situation in which two market participants would each be better off if they exchanged their respective property. It captures the idea of a potential welfare benefit that could be realized if trade was allowed and possible. Generally, in this release the relevant property will be securities and cash. Market participants that post the orders that are traded against would also benefit from realizing additional gains from trade.

Changes in the NBBO could also affect other Commission and SRO rules. Market participants should benefit from these changes independently of any benefits from the decentralized consolidation model.

The Commission recognizes that there are costs to expanding the content of consolidated market data. They include costs to new competing consolidators related to upgrading existing infrastructure in order to handle the dissemination of the increased message traffic; costs relating to upgrading software and trading systems that consume consolidated market data; costs relating to market participants receiving consolidated market data from technological investments required to handle increased content and message traffic.[1594] Expanding consolidated market data will also result in transfers among various market participants, including transfers from the current beneficiaries of asymmetric information associated with the uneven distribution of market data to market participants who currently do not have access to the additional information contained in proprietary DOB products and other proprietary products. SROs will have costs associated with the dissemination of data content underlying consolidated market data.

With respect to the introduction of the decentralized consolidation model, the Commission believes that the risk of too few competing consolidators operating in the market and precluding any of the potential benefits from materializing is low, and in any event, certain benefits from opening up the market to competitive forces will materialize even with few competing consolidators because the market will now be open to new entrants, i.e., benefits from the threat of entry. The potential economic benefits of the decentralized consolidation model will include a reduction in the latency (as measured at the location of market participants using the data) and content differential that exists between SIP data and proprietary data feeds,

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[1594] See infra Section V.C.1 for a complete discussion of related costs.

improvements in innovation and efficiency in the consolidated market data delivery space, and an increase in market resiliency. Moreover, because today's market participants need to subscribe to both the exclusive SIPs and proprietary data feeds to receive the same content that will be included in consolidated market data, the Commission expects the fees for consolidated market data will likely be lower than fees that market participants pay for equivalent data today.[1595] Finally, subscribers choosing to receive a subset of consolidated market data will likely pay the same or lower fees than they do today for equivalent data, depending on the fee schedule of the effective market system plan(s).

At the same time, the introduction of the decentralized consolidation model will impose direct costs on potential competing consolidators and SROs.[1596] Potential competing consolidators (such as SROs, exclusive SIPs, and current market data aggregators) will incur registration and compliance costs and implementation and incremental infrastructure costs.[1597] SROs will incur costs as part of their SRO functions, which include costs to file amendments to the effective national market system plan(s) and to collect and disseminate the data content underlying new elements of consolidated market data to competing consolidators.

The final rule will also impose indirect costs on the existing exclusive SIPs, certain market participants and investors, and on SROs.[1598] The existing exclusive SIPs will incur a loss in revenue as they lose their role as the exclusive distributors of consolidated market data. The SROs might incur indirect costs depending on how they choose to provide the data content

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1595 See infra Section V.C.2(b) for an analysis of the impact on data fees.

1596 See infra Section V.C.2(d).

1597 Many of the potential competing consolidators have already invested in this infrastructure for the existing business services that they provide (e.g., proprietary data aggregation services), which may reduce their implementation costs.

1598 See infra Section V.C.2(d) for a discussion of the related costs.

underlying consolidated market data. Finally, certain market participants will incur direct or indirect implementation costs and switching costs to use the consolidated market data products.

The Commission believes that the interaction of expanding consolidated market data and implementing a decentralized consolidation model together should produce some benefits, including less expensive alternatives to proprietary DOB products for market participants; potential new entrants into the broker-dealer, market making, and other latency sensitive trading businesses;[1599] expansion of business opportunities for market data aggregators; improved regulatory oversight from the Consolidated Audit Trail;[1600] and enhancements to the quality of service provided by data vendors. Further, as noted above, the Commission believes that the adopted rule will facilitate best execution and reduce information asymmetries. These changes might impose certain costs, such as potentially lower revenues for SROs; potentially higher costs for the implementation of the Consolidated Audit Trail; potentially higher costs for certain market data vendors.[1601] Some of these benefits and costs will result from transfers among various market participants.

On balance, the amendments are necessary and appropriate in the public interest and do not impose a burden on competition not necessary or appropriate in furtherance of the purposes of the Exchange Act.

1599 This includes the indirect benefits of improved competition in the executing broker-dealer business and potential increases in market liquidity from additional market makers.

1600 The expanded content of core data will improve the completeness and accessibility of Consolidated Audit Trail Data, which will facilitate more efficient regulatory activities using Consolidated Audit Trail Data. See infra Section V.C.4(c)(ii).

1601 See infra Section V.C.4 for additional discussion of the related costs.

Wherever possible, the Commission has quantified the likely economic effects of the adopted rules. The Commission is providing both a qualitative assessment and quantified estimates of the economic effects of the adopted rule where feasible. The Commission has incorporated data and other information provided by commenters to assist it in the analysis of the economic effects of the adopted rules.[1602]

2. Market Failures

The Commission is amending Rules 600 and 603 and adopting new Rule 614 of Regulation NMS under the Exchange Act to increase the availability and improve the dissemination of information regarding quotations for and transactions in NMS stocks to market participants. First, the Commission is defining the terms "consolidated market data," "consolidated market data product," "core data," "regulatory data," "administrative data," and "self-regulatory organization-specific program data," and enhancing the content of core data to include certain odd-lot quote information, certain depth of book data, and information on orders participating in auctions.[1603] Second, the Commission is introducing a decentralized consolidation model whereby competing consolidators will assume responsibility for the collection, consolidation, and dissemination functions currently performed by the exclusive

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1602 As explained in more detail below, because in certain circumstances the Commission may not have, and in certain cases cannot reasonably obtain, data that may inform the Commission on certain economic effects, the Commission is unable to quantify certain economic effects. Further, in certain circumstances, it may not be practicable to quantify the economic effects due to the number and type of assumptions necessary, which render any such quantification unreliable.

1603 See supra Section II.